4.Implement histogram equalization on the given image and compare it with the original image using Open CV

1. Aim

To write and execute a Python program to perform histogram equalization on a given image and compare the equalized image with the original image using OpenCV.

2. Procedure

1. Import the OpenCV library using import cv2.
2. Read the input image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Convert the original image into grayscale using cv2.cvtColor().
5. Apply histogram equalization using cv2.equalizeHist().
6. Display the original grayscale image and the equalized image using cv2.imshow().
7. Compare both images and observe the improvement in contrast.
8. Press any key to close the image windows.

3. Code

```
import cv2


# Read the image
img = cv2.imread("img1.jpg")


if img is None:
    print("Error: Image not found!")
else:
    # Convert image to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)


    # Apply histogram equalization
    equalized = cv2.equalizeHist(gray)


    # Display original and equalized images
    cv2.imshow("Original Image", gray)
```

```
cv2.imshow("Histogram Equalized Image", equalized)


# Save the equalized image
cv2.imwrite("equalized_image.jpg", equalized)


cv2.waitKey(0)
cv2.destroyAllWindows()4.Sample input:
```



5.sample output:



4. Result

Thus, histogram equalization was successfully performed on the given image using OpenCV. The original grayscale image and the equalized image were compared, and the equalized image showed improved contrast.